

Tom de Groot

Frontier Evaluation & Formal Verification

Grijpskerk, NL

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github.com/Tomodovodoo

x.com/Tomodovodoo

TECHNICAL STACK

ML/Agents: TensorFlow, n8n, OpenClaw, Slurm

Formal: Lean 4, mathlib, Lake, CI/CD for proofs, auto-formalization.

Tools: Git, GitHub, Linux/HPC, VS Code, Docker, Numba/Njit

Software Languages: Python, Lean.

Spoken Languages: Dutch, English.

EDUCATION

BSc Chemistry

University of Groningen

2021 - 2025

Track: Chemistry of Life

Thesis: Evaluating and Optimizing NMR Prediction with Graph Neural Networks (Chem-AI)

AWARDS

Chess.com Member of the Year 2021

Harmonic Rising Mathematician Grant (Details on request)

ABOUT ME

Hi, I'm Tom. I'm a BSc Chemistry graduate from the University of Groningen (Life Sciences track). My thesis in the Pollice Group focused on optimizing NMR prediction with GNNs. Post-graduation, I pivoted to mathematics and formal methods, becoming an active contributor to the **PNT+** Lean formalization project. I've built a strong foundation in student mathematics and programming, which I use to design and evaluate AI systems, their output, and create frameworks for them.

Goals! Focusing on frontier model capability work, stress-testing, and evaluation design. My current goal is to use agentic systems and formal verification (Lean 4) to contribute to open science in mathematics.

SELECTED RESEARCH & FORMALIZATION

Prime Number Theorem And (PNT+)

Late 2025 - Present

Active contributor to the Lean 4 formalization project led by Alex Kontorovich and Terence Tao. Using an AI system called Aristotle, I have formalized multiple intermediate theorems, and I am a recent maintainer of the Aristotle branch alongside Pietro Monticone.

My current focus herein is the scaling up of Aristotle and other agentic systems to formalize more mathematics. Merged PRs

First Proof Problem 9 (Lean-verified Solution)

Feb 2026

Produced the only credible formalized solution during the 7 day challenge, recognized by among others **Kevin Buzzard** and **Daniel Litt**, who gave it the context of "gold standard of LLM-aided mathematics." Created an autonomous agent that worked on the informal proof, and orchestrated multiple LLM agents to formalize it in Lean 4. View Repository

ENGINEERING PROJECTS

BaanSolliciteren (Agentic Job Automation)

Engineered a semi-agentic system to automate my job search and its documentation. Built a custom **Gmail MCP server** (Model Context Protocol) to enable real-time status tracking and ranking of vacancies.

Implemented HTML-to-JSON schema parsing and automated LLM-based fit-scoring. Developed logic to cross-reference vacancy descriptions against personal descriptors, significantly reducing manual review time.

Technical Collection & Visualizations

Multi-year repository of technical interests and explorations: including from-scratch audio-to-text implementations and interactive visualizations of high-dimensional datasets (e.g., vaccine distribution and water quality metrics). Focused on extracting insights from real-world data. GitHub

PROFESSIONAL EXPERIENCE

Transcom (for Odido)

2024

Demonstrated reliable performance in customer service, maintaining consistent top-tier customer satisfaction ratings and learning to explain at every level.

Teamspot (Regional Director)

2022 - 2024

Coordinated regional school projects and taught teachers, performed technical testing/data entry within an Agile/Jira environment, alongside in-depth bug testing and general responsibilities.

Chess.com

2022

Moderated and proctored international chess events; recognized as **Member of the Year 2021** for community ownership and chess coaching.